Supplementary Material for "Minimum Viable State: What Adam Must Carry Across a Preemption"

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S1 Results at Each Model Size

Table S1 gives the main measure, peak excess held-out loss divided by the seed band, at each of the three model sizes of the main sweep, pooled over the four migration points. Brackets are 95% seed-bootstrap intervals; the second column of each pair counts the runs that never left the band.

TABLE S1
Peak excess loss over the seed band, by model size.

	0.8M parameters		10.7M parameters		25.3M parameters	
Strategy	Peak/ε	StR = 0	Peak/ε	StR = 0	Peak/ε	StR = 0
`full`	0.00 [0.00, 0.00]	12/12	0.00 [0.00, 0.00]	12/12	0.00 [0.00, 0.00]	12/12
`w_mq8_vlog8`	0.07 [0.04, 0.08]	12/12	0.02 [0.01, 0.03]	12/12	0.09 [0.08, 0.11]	12/12
`w_m_vr1`	0.16 [0.15, 0.17]	12/12	0.10 [0.09, 0.11]	12/12	0.17 [0.14, 0.22]	12/12
`w_v`	0.20 [0.18, 0.21]	12/12	0.30 [0.20, 0.39]	11/12	0.29 [0.28, 0.30]	12/12
`w_vsvd4`	0.22 [0.21, 0.23]	12/12	0.31 [0.22, 0.39]	11/12	0.33 [0.30, 0.36]	12/12
`w_emb_v`	0.28 [0.26, 0.29]	12/12	0.31 [0.22, 0.40]	11/12	0.31 [0.29, 0.35]	12/12
`w_vr1`	0.22 [0.20, 0.22]	12/12	0.31 [0.21, 0.39]	11/12	0.31 [0.30, 0.34]	12/12
`wbf16_vr1`	0.26 [0.24, 0.29]	12/12	0.31 [0.22, 0.39]	11/12	0.31 [0.29, 0.33]	12/12
`w_m`	42.10 [40.63, 43.07]	0/12	102.63 [76.22, 147.32]	0/12	69.26 [66.83, 72.29]	0/12
`w_zero`	2.51 [2.11, 2.86]	3/12	9.58 [8.85, 10.79]	1/12	11.40 [10.30, 12.13]	3/12
`w_fresh`	1.08 [1.05, 1.12]	7/12	4.11 [3.18, 4.96]	6/12	4.38 [3.57, 5.76]	5/12
`w_rewarm`	0.71 [0.60, 0.83]	12/12	0.89 [0.87, 0.92]	9/12	0.83 [0.79, 0.85]	11/12
`w_warmv`	0.22 [0.20, 0.23]	12/12	0.34 [0.29, 0.42]	11/12	0.40 [0.36, 0.46]	10/12
`w_m_warm`	0.14 [0.12, 0.16]	12/12	0.38 [0.21, 0.63]	11/12	0.55 [0.45, 0.71]	9/12
`w_warm`	3.72 [3.35, 4.09]	3/12	9.23 [8.66, 10.10]	4/12	8.44 [7.76, 9.63]	4/12

S2 Peaks by Migration Point

Table S2 gives the numbers behind the heat map of the main text for the largest model: the peak excess held-out loss in mnats at each migration point, with 95% seed-bootstrap intervals, and the seed band of each point in the first row.

S3 Latency Breakdown

Table S3 gives the stage times behind the latency figure of the main text. Each strategy was run for 20 timed rounds after three untimed ones, and no other job used the device during timing.

S4 Cross-Device Gaps Over Time

Fig. S1 plots the held-out loss of the `full` payload resumed on the CPU and on the integrated GPU against the same payload resumed on the discrete GPU, in single precision without dropout. The grey band is the seed band of the same model and migration point.

S5 The Larger Model

Table S4 gives the 85M-parameter model in the format of the main table: 3 seeds, migrations at 10% and 60% of the schedule, five strategies.

TABLE S2
Peak excess held-out loss in mnats by migration point (25.3M parameters).

Strategy	Step 200 (10%)	Step 600 (30%)	Step 1200 (60%)	Step 1700 (85%)
ε (seed band)	24.8	22.7	18.4	16.4
`w_mq8_vlog8`	6.1 [5.1, 6.7]	2.4 [0.8, 5.0]	0.2 [0.1, 0.2]	0.1 [0.1, 0.1]
`w_m_vr1`	10.5 [8.3, 14.1]	4.5 [3.6, 6.2]	0.8 [0.3, 1.7]	0.4 [0.2, 0.7]
`w_v`	18.5 [16.1, 20.1]	7.0 [5.5, 8.4]	1.7 [0.9, 2.1]	0.1 [-0.1, 0.3]
`w_vsvd4`	22.5 [17.7, 28.1]	7.5 [5.5, 8.7]	1.6 [1.3, 2.0]	0.1 [-0.1, 0.3]
`w_emb_v`	21.6 [17.8, 25.1]	6.1 [4.5, 7.1]	1.8 [1.4, 2.2]	0.2 [0.0, 0.5]
`w_vr1`	21.6 [18.3, 24.4]	6.3 [5.0, 7.9]	1.7 [1.4, 2.2]	0.1 [-0.1, 0.3]
`wbf16_vr1`	21.7 [20.1, 22.7]	6.3 [4.5, 7.3]	1.8 [1.4, 2.2]	0.2 [-0.1, 0.4]
`w_m`	1539.2 [1507.7, 1593.9]	2139.7 [2051.3, 2279.0]	1983.2 [1816.9, 2114.7]	214.5 [204.2, 222.8]
`w_zero`	750.1 [690.2, 827.6]	285.1 [253.0, 343.0]	40.5 [29.5, 55.5]	9.3 [6.6, 11.5]
`w_fresh`	347.9 [267.4, 496.2]	60.9 [42.1, 74.9]	12.1 [6.6, 20.3]	2.0 [1.1, 3.6]
`w_rewarm`	56.9 [52.4, 59.8]	18.7 [15.6, 21.1]	2.0 [1.5, 2.6]	1.5 [-0.1, 3.5]
`w_warmv`	28.5 [24.6, 35.1]	7.4 [6.8, 8.0]	1.9 [1.9, 1.9]	0.4 [-0.0, 0.7]
`w_m_warm`	39.2 [27.8, 54.2]	11.1 [8.4, 13.2]	1.3 [1.1, 1.5]	0.7 [0.7, 0.8]
`w_warm`	663.1 [617.9, 739.1]	130.0 [92.3, 161.8]	21.7 [12.8, 26.8]	1.2 [0.9, 1.5]

TABLE S3
Latency of one migration in milliseconds on the discrete GPU (25.3M parameters, median of 20 rounds; brackets are the 5th and 95th percentiles).

		Drain (source)					Restore (destination)				
Strategy	MB	encode	to host	serialise	write	total	read	to device	decode	rebuild	total
`full`	304.1	0.5	24.7	276.5	244.7	579.8 [523.2, 635.9]	184.5	39.6	0.8	9.8	427.0 [338.4, 531.4]
`w_mq8_vlog8`	153.3	89.9	19.6	150.0	167.1	441.9 [372.0, 501.9]	95.0	25.8	26.6	9.5	248.4 [209.3, 302.3]
`w_m_vr1`	203.0	2.7	16.9	170.6	204.3	417.1 [372.0, 483.2]	107.2	28.1	6.2	8.0	245.6 [208.2, 361.9]
`w_v`	202.7	0.4	31.7	196.6	195.8	426.9 [387.1, 497.8]	105.5	40.3	0.4	11.7	257.4 [220.8, 361.6]
`w_vsvd4`	102.5	839.1	9.0	86.4	116.5	1052.4 [1026.6, 1070.9]	59.4	14.4	2.4	8.0	131.2 [123.0, 142.0]
`w_emb_v`	102.3	3.2	17.0	83.5	114.6	224.3 [210.7, 248.1]	60.0	26.0	7.4	11.9	155.0 [133.5, 229.9]
`w_vr1`	101.7	3.0	15.0	93.7	105.5	219.4 [206.6, 258.7]	61.3	21.2	7.4	11.3	157.4 [138.1, 228.1]
`wbf16_vr1`	51.0	4.2	9.0	50.1	61.6	127.1 [110.1, 153.8]	40.1	14.8	9.4	11.1	105.9 [94.7, 163.2]
`w_m`	202.7	0.3	16.6	168.9	177.6	367.6 [355.6, 451.3]	101.8	27.5	0.4	10.8	215.2 [201.7, 337.8]
`w_zero`	101.4	0.2	9.0	84.7	115.9	222.6 [204.0, 240.9]	61.0	15.0	0.3	14.7	147.0 [125.7, 213.5]
`w_fresh`	101.4	0.2	16.3	108.6	115.6	246.7 [224.8, 275.7]	55.9	22.1	0.3	15.6	142.0 [126.6, 221.6]
`w_rewarm`	101.4	0.2	20.8	97.3	111.3	228.4 [205.4, 265.3]	54.6	17.7	0.2	18.2	145.6 [120.0, 215.8]
`w_warmv`	101.4	0.2	8.7	97.2	112.2	221.3 [197.5, 238.7]	60.3	11.9	0.2	5620.8	5738.4 [5703.0, 5811.3]
`w_m_warm`	202.7	0.4	16.6	189.1	180.6	387.7 [365.6, 444.8]	103.7	24.2	0.4	5524.8	5750.6 [5437.3, 5932.8]
`w_warm`	101.4	0.2	8.6	91.3	106.1	208.7 [188.3, 219.7]	58.0	12.1	0.2	5627.5	5742.1 [5685.1, 5799.6]

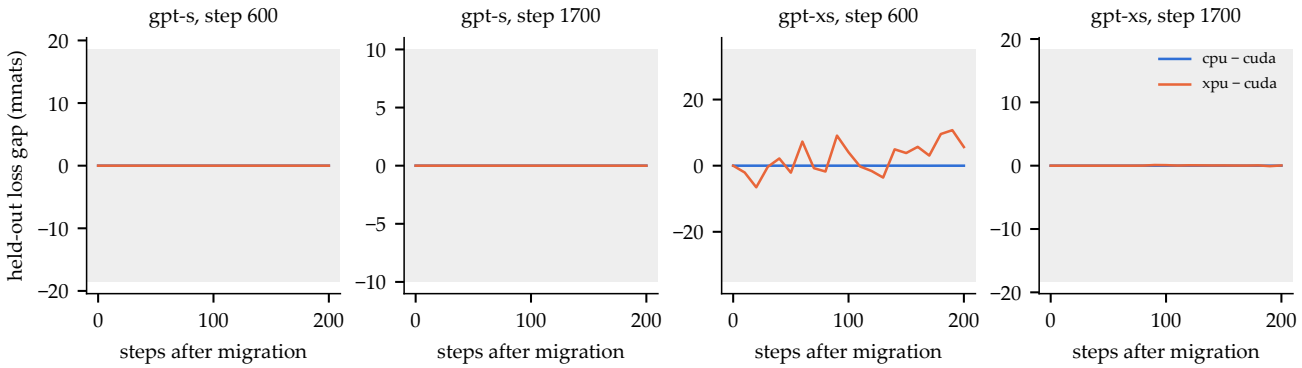


Fig. S1. Held-out loss gap between devices after resuming the same full payload.

S6 Preemption and Resizing with Multiple Ranks

Table S5 repeats the end-to-end experiment of the main text with CPU ranks on one host, changing the number of ranks at each preemption: two ranks shrink to one, and a later preemption grows the job back to two. The run ended with a held-out loss 13.7 mnats from the uninterrupted run, and mean time to recovery was 6.788 s. It took 930 s of wall-clock time against 996 s for its control, so the cost of the three interruptions is below the timing noise of CPU runs on this machine.

TABLE S4
Cost of each strategy on the 85M-parameter model. Columns as in the main table.

Strategy	Payload %	Peak (mnats)	Peak/ε	Regret (nat-steps)	Steps lost	StR $= 0$	StR	$> W$
`full`	100.0	0.0 [0.0, 0.0]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.0 [0.0, 0.0]	6/6	–	0
`w_mq8_vlog8`	50.4	3.8 [2.6, 5.0]	0.17 [0.12, 0.22]	0.19 [0.09, 0.29]	0.1 [0.1, 0.2]	6/6	–	0
`w_vr1`	33.4	9.7 [9.1, 10.8]	0.45 [0.42, 0.50]	0.92 [0.80, 1.03]	4.7 [4.1, 5.2]	6/6	–	0
`w_zero`	33.3	384.7 [356.4, 436.0]	17.31 [16.04, 19.60]	31.06 [29.47, 33.57]	94.3 [90.0, 97.2]	0/6	30	3
`w_fresh`	33.3	197.0 [157.3, 219.5]	8.81 [7.04, 9.83]	12.70 [10.20, 15.08]	40.8 [32.7, 48.0]	3/6	–	3

TABLE S5
Injected preemptions and a crash with a change of world size (0.8M parameters, CPU ranks on one host). Times in seconds unless marked.

Event	Step	World	Steps redone	Payload (MB)	Drain	Restart	Init	Fetch (ms)	Rebuild (ms)	Downtime
preempt	150	2→1	0	3.364	1.367	1.744	1.798	10.61	9.58	6.378
crash	300	1→1	1	5.033	–	2.023	2.158	18.9	3.15	6.82
preempt	450	1→2	0	3.364	1.371	1.758	2.588	7.68	6.7	7.165

S7 Reproduction

Each table and figure is rebuilt from the run directories by one command, `python scripts/paper.py`; the experiments themselves are launched with `scripts/run_experiment.py`, `scripts/xdevice.py`, `scripts/latency.py`, `scripts/trace.py`, `scripts/diagnose.py` and `scripts/chaos.py` and the configuration files under `configs/experiments`. Every run directory holds a manifest with the commit, the resolved configuration, the device and library versions, the measured payload size and the median step time.